



PHY101-PHYSICS
(SOSUBJECTIVE)
FROM MIDTERM PAPERS
LECTURE (1-22)



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CS619 & CS519

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1. A generator supplies 100V to the primary coil of a transformer . the primary has 50 turns the secondary voltage is :
 - ☒ 1000V
 - ☐ 500V
 - ☐ 250V
 - ☐ 100V
2. Polarization experiments provide evidence that light is:
 - ☐ A longitude wave
 - ☒ A transverse wave
 - ☐ Some type of wave
 - ☐ Nearly monochromatic
3. The units of the plank constant h are those of :
 - ☒ Energy
 - ☐ Angular momentum
 - ☐ Momentum
 - ☐ Frequency
4. We desire to make an LC current that ascllates 100 Hz using an inductance of 2.5H . we also need a capacitance of
 - ☐ 1 F
 - ☐ 1mF
 - ☒ 1uf
 - ☐ 100uf
5. If energy of photon is E, it is equvalent to mass:
 - ☐ E/c^2
 - ☐ Ec^2
 - ☐ C^2/E
 - ☒ Ec
6. The wave length of light beam A is twice the wavelength of light beam B. the energy of a photon in beam A is:
 - ☒ Half the energy of photon in beam B
 - ☐ One-fourth the energy of a photon in beam B
 - ☐ Equal to the energy of a photon in beam B

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✚ Twice the energy of photon in beam B

7. The reaction of an induction at 50 Hz is $10\ \Omega$, its reactance at 100Hz becomes:

✚ $20\ \Omega$

✚ $5\ \Omega$

✚ $2.5\ \Omega$

✚ $1\ \Omega$

8. The wavelength of red light is 700 nm. Its frequency is

✚ 4.29×10^3 Hertz

✚ 4.29×10^5 Hertz

✚ 4.29×10^4 Hertz OR 4.29×10^{14} Hertz

✚ 4.29×10^2 Hertz

9. $2d\sin\theta = m\lambda$ is called:

✚ Laplace's equation

✚ Reflection equation

✚ Refraction equation

✚ Bragg equation

10. A step-down transformer is used to :

✚ Increase the power

✚ Decrease the power

✚ Increase the voltage

✚ Decrease the voltage

11. The focal length of a spherical mirror is N times its radius of curvature where N is:

✚ $\frac{1}{4}$

✚ $\frac{1}{2}$

✚ 1

✚ 2

12. The appearance of colors in film is due to:

✚ Interference

✚ Diffraction

✚ Dispersion

✚ Polarization

13. Bending of light around the edges of an obstacle called:

✚ Refraction

✚ Polarization

✚ Interference

✚ Diffraction

14. In constructing a thermometer it is NECESSARY to use a substance that:

✚ Expands linearly with rising temperature

✚ Will not freeze

✚ Will not boil

✚ Undergoes some change when heated or cooled

15. In Young's double slit experiment, the position of the bright fringe is given by:

✚ $Y_m = mU\lambda d$

✚ $Y_m = mU\gamma d$

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✚ $Y_m = m\lambda d/L$

✚ $Y_m = mLd\lambda$

16. Constant-volume gas thermometers using different gases all indicate nearly the same temperature when in contact with the same object if :

✚ The volume are all extremely large

✚ The volume are all the same

✚ The pressure are all extremely large

✚ The particle concentration are all extremely small

17. In meta stable state electrons reside:

✚ 10^{-8} sec

✚ 10^{-5} sec

✚ 10^{-3} sec

✚ 10^{-13} sec

18. Following are the ways by which light can interact with matter , EXCEPT:

✚ Emission

✚ Interference

✚ Absorption

✚ Transmission

19. In RL series circuit phase angle is given by:

✚ $\tan^{-1}(\omega R/L)$

✚ $\tan^{-1}(R/\omega L)$

✚ $\tan^{-1}(\omega L/R)$

✚ $\tan^{-1}(\omega R/L)$

20. The theoretical upper limit for the frequency of electromagnetic wave is :

✚ Just slightly greater than of red light

✚ Just slightly less than that blue light

✚ The greatest x-ray frequency

✚ There is no upper limit

21. The plants look green. It is because they have chlorophyll in them .chlorophyll make leaves green to absorb_____:

✚ Green Only

✚ Red only

✚ Blue only

✚ Red and blue both

22. Electric energy is measured:

✚ Kilowatt hour

✚ Kilowatt

✚ Horsepower

✚ Watt

23. The special theory of relativity is applicable to the object moving with maximum velocity equal to:

✚ Speed of light

✚ More than speed of light

✚ Less than speed of light

✚ Double the speed of light

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24. In some movies you sometimes see an actor looking in a mirror and you can see his face in the mirror. during the filming of the scene what does the actor see in the mirror?

- ☐ His face
- ☐ Your face
- ☒ The movies cameras
- ☐ The director's face

25. The units of Stefan boltzomen constant are_____:

- ☐ $Wm^{-2}K^{-4}$
- ☐ $W^{-2}m^{-2}K^{-4}$
- ☒ $Wm^{-2}Kc^{-4}$
- ☐ $Wm^{-2}K^{-4}$

26. Two sources of light are said to be coherent if the waves produced by them have the same:

- ☐ Wavelength
- ☐ Amplitude
- ☒ Wavelength and amplitude
- ☐ Wavelength and constant phase difference

27. Antiparticle of electron is:

- ☐ Proton
- ☐ Photon
- ☐ Neutron
- ☒ Positron

28. The goal of all scientific inquiry (or scientific method) is:

- ☒ Predicting natural events based
- ☐ Predicting artificial events based
- ☐ Predicting immortal events based
- ☐ Predicting normal events based

29. The law of conservation of momentum applies to a system of colliding objects only if:

- ☒ The net external Impulse is zero
- ☐ The next internal Impulse is zero
- ☐ The net external Impulse is one
- ☐ The next internal Impulse is one

30. Work has dimensions like:

- ☒ Power
- ☐ Energy
- ☐ Force
- ☐ Velocity

31. The work done by gravity during the descent of a projectile:

- ☒ Is positive
- ☐ Is negative

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- ☐ Is natural
- ☐ None
32. An object moves in a circle at constant speed. The work done by the centripetal force is zero because:
- ☒ The centripetal force is perpendicular to the velocity
- ☐ The centripetal force is equal to the velocity
- ☐ The centripetal force is not equal to the velocity
- ☐ The centripetal velocity is perpendicular to the force
33. The dimensional units of ratio of work and power is:
- ☒ T
- ☐ N
- ☐ Ns^{-1}
- ☐ m s^{-1}
34. A heavy particle moving with 5 m/s suffers an elastic collision with a light particle at rest. After collision, the velocity of light particle will be:
- ☒ 5 m/s
- ☐ 6 m/s
- ☐ 4 m/s
- ☐ 2 m/s
35. Which of the following statement/s is / are true?
- ☒ 1 only
- ☐ 2 only
- ☐ 3 only
- ☐ 4 only
36. The law of inertia was firstly formulated by:
- ☒ Galileo
- ☐ Pascal
- ☐ Newton
- ☐ Charles
37. The product of force and time is called change in:
- ☒ Momentum
- ☐ Force
- ☐ Velocity
- ☐ None
- OR
- ☐ Torque
- ☐ Acceleration
- ☐ Force
- ☒ Impulse
38. It is easier to walk on the concrete road than on ice because:

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☒ The friction between the ice and the feet is lees

☐ The friction between the feet and the ice is lees

☐ The friction between the ice and the feet is more

☐ The friction between the ice and the feet is high

39. Light year is a unit of:

☒ Distance

☐ Displacements

☐ Acceleration

☐ Force

40. A force of 100 N acts upon a body for five seconds. What will be the change in momentum? _____.

☒ 500Ns

☐ 400 Ns

☐ 300 Ns

☐ 200 Ns

41. Add two vectors of length 4 m & 5 m but their orientation is not known. The length after addition of these two vectors will be:

☒ Between 9m and 1m

☐ Between 8m and 3m

☐ Between 7m and 4m

☐ Between 1m and 9m

42. Example of physical quantity is:

☒ Length

☐ Width

☐ Height

☐ Strength

43. An inelastic collision is one in which:

☒ kinetic energy

☐ Potential energy

☐ Gravitation energy

☐ Heat energy

44. Acceleration in a body is always produced in the direction of:

☒ Velocity

☐ Force

☐ Distance

☐ Displacement

45. Ali wants to lift a mass of 7.5 kg with constant velocity by a rope that passes through a frictionless pulley which is attached to the ceiling of room. Calculate the tension in the rope by neglecting the mass of the rope.

☐ 0.70N

☒ 0.75N

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☐ 0.65 N

☐ 0.60 N

46. What is the angle of projection of projectile, for which its maximum height and horizontal range are equal?

☒ 76°

☐ 77°

☐ 66°

☐ 56°

47. The SI standard of time is based on:

☐ none of these

☒ second

☐ m/s

☐ N

48. A net torque applied to a rigid object always tends to produce:

☒ Rotational equilibrium

☐ Linear acceleration

☐ Irrotational equilibrium

☐ None

49. An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

☐ 12.6 rad/s

☒ 2.0 rad/s

☐ 3.0 rad/s

☐ 4.0 rad/s

50. In simple harmonic motion, the restoring force must be proportional to the:

☐ Velocity

☐ Distance

☐ Force

☒ Displacement

51. Mercury is a convenient liquid to use in a barometer because: it is a metal

☒ it has a high density

☐ it has a low density

☐ it has a less density

☐ it has a more than density

52. The units of the electric field are:

☒ J/m

☐ J/(C·m)

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- ☐ N/s
- ☐ m/s

53. A farad is the same as a

- ☒ J/V
- ☐ V/J
- ☐ J/m
- ☐ N/s

54. We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

- ☐ 1 F
- ☒ 100 μ F
- ☐ 100 kF
- ☐ 100 gF

55. Which of the following statements is NOT TRUE about electromagnetic?

- ☐ Electromagnetic waves satisfy the Maswell's Equation.
- ☒ The electromagnetic radiation from a burning candle is unpolarized.
- ☐ All of the given options
- ☐ None of these

56. Radio waves and light waves are _____.

- ☐ Longitudinal waves
- ☐ Transverse waves
- ☐ Electromagnetic waves
- ☒ Electromagnetic and transverse both

57. Wien's Law states that, $\lambda_{max} =$ _____K.

- ☐ 2.90×10^{-3} kg
- ☐ 2.90×10^{-3} mg
- ☐ 2.90×10^{-3} cg
- ☒ 2.90×10^{-3} m

58. Interference of light is evidence that:

- ☐ the speed of light is very large
- ☐ the speed of light is very short
- ☐ light is not a wave phenomenon
- ☒ light is a wave phenomenon

59. Fahrenheit and Kelvin scales agree numerically at a reading of:

- ☒ -40

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- ☐ 0
- ☐ 50
- ☐ 56

60. According to the theory of relativity:

- ☒ moving clocks run fast
- ☐ energy is not conserved in high-speed collisions
- ☐ energy is conserved in high-speed collisions
- ☐ moving clocks run slow

61. Light from a stationary spaceship is observed, and then the spaceship moves directly away from the observer at high speed while still emitting the light.

- ☐ As a result, the light seen by the observer has:
- ☐ higher frequency and a longer wavelength than before
- ☒ lower frequency and a shorter wavelength than before
- ☐ none of these

62. How fast should you move away from a 6.0×10^{14} Hz light source to observe waves with a frequency of 4.0×10^{14} Hz?

- ☐ 20c
- ☐ 21c
- ☐ 39c
- ☒ 38c

63. The quantum number n is most closely associated with what property of the electron in a hydrogen atom?

- ☒ Energy
- ☐ Orbital angular momentum
- ☐ Power
- ☐ Force

64. As the wavelength of a wave in a uniform medium increases, its speed will_____.

- ☐ Decrease
- ☐ Increase
- ☐ Not same
- ☒ Remain the same

65. The lowest tone produced by a certain organ comes from a 3.0-m pipe with both ends open. If the speed of sound is 340m/s, the frequency of this tone is approximately:

- ☐ 7Hz
- ☐ 95Hz
- ☐ 105Hz

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✚ 57 Hz

66. To raise the pitch of a certain piano string, the piano tuner:

- ✚ loosens the string
- ✚ tightens the string
- ✚ harder the string
- ✚ soft the string

67. An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

- ✚ 12.6 rad/s
- ✚ 1.57 rad/s
- ✚ 1.95 rad/s
- ✚ 15.78 rad/s

68. For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

- ✚ the center of mass
- ✚ the center of circle
- ✚ not same point
- ✚ the same point

69. Ten seconds after an electric fan is turned on, the fan rotates at 300 rev/min. Its average angular acceleration is:

- ✚ 3.14 rad/s²
- ✚ 30 rad/s²
- ✚ 40 rad/s²
- ✚ 48 rad/s²

70. A 4.0-N puck is traveling at 3.0m/s. It strikes a 8.0-N puck, which is stationary. The two pucks stick together. Their common final speed is:

- ✚ 1.0m/s
- ✚ 1.5m/s
- ✚ 2 m/s
- ✚ 4 m/s

71. An object moving in a circle at constant speed:

- ✚ must have only one force acting on it
- ✚ has an acceleration of variable on it
- ✚ none
- ✚ has an acceleration of constant magnitude

72. A plane traveling north at 200m/s turns and then travels south at 200m/s. The change in its velocity is:

- ✚ 400m/s north

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☒ 400m/s south

☐ 400 m/s east

☐ 400 m/s west

73. At time $t = 0$ a car has a velocity of 16 m/s. It slows down with an acceleration given by $-0.50t$, in m/s² for t in seconds. It stops at $t =$

☐ 64 s

☒ 8.0 s

☐ 50 s

☐ 40 s

74. 1 mi is equivalent to 1609 m so 55 mph is: 15 m/s

☒ 25 m/s

☐ 45 m/s

☐ 50 m/s

☐ 60 m/s

75. The number of significant figures in 0.00150 is:

☐ 5

☐ 4

☐ 6

☒ 3

76. For a body to be in equilibrium under the combined action of several forces:

☐ All the forces must be applied at the same point

☐ None of these

☐ Both are correct

☒ any two of these forces must be balanced by a third force

77. A bucket of water is pushed from left to right with increasing speed across a horizontal surface. Consider the pressure at two points at the same level in the water.

☐ It is the same

☐ It is not same

☐ It is lower at the point on the right

☒ It is higher at the point on the left

78. An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

☐ 200 Hz

☐ 300 Hz

☐ 400 Hz

☒ 600 Hz

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79. Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by

☒ $C_1 C_2 / (C_1 + C_2)$

☐ $(C_1 - C_2) / C_1 C_2$

☐ $(C_1 * C_2) / C_1 C_2$

☐ $(C_1 / C_2) / C_1 C_2$

80. If the potential difference across a resistor is doubled:

☒ Only the current is doubled

☐ Only the current is halved

☐ Only the current is single

☐ Only the current is triple

81. By using only two resistors, R1 and R2, a student is able to obtain resistances of 3Ω, 4Ω, 12Ω, and 16 Ω. The values of R1 and R2 (in ohms) are:

☐ 3, 9

☐ 5, 15

☐ 6, 18

☒ 4, 12

82. Faraday's law states that an induced emf is proportional to: the rate of change of the electric field

☒ the rate of change of the magnetic flux

☐ the rate of not change of the magnetic flux

☐ None of these

☐ Both are correct

83. A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

☒ 1000V

☐ 500V

☐ 600V

☐ 2000V

84. Which of the following electromagnetic radiations has photons with the greatest energy?

☐ blue light

☐ white light

☐ green light

☒ x rays

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85. A virtual image is one:

- ☐ toward which light rays converge but do not pass through
- ☐ Both are correct
- ☐ None of these
- ☒ from which light rays diverge as they pass through

86. What is the unit of magnification factor?

- ☐ meter.
- ☐ Kelvin
- ☐ Pascal
- ☒ no unit

87. During an adiabatic process an object does 100 J of work and its temperature decreases by 5K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is.

- ☐ 100 J/K
- ☐ 150 J/K
- ☐ 200 J/K
- ☒ 15 J/K

88. An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

- ☒ a change in entropy
- ☐ a decrease of internal energy
- ☐ not change in entropy
- ☐ a increase of external energy

89. . The Stern-Gerlach experiment makes use of:

- ☐ a strong uniform magnetic field
- ☐ a weak uniform magnetic field
- ☐ a weak non-uniform magnetic field
- ☒ a strong non-uniform magnetic field

90. A large collection of nuclei are undergoing alpha decay. The rate of decay at any instant is proportional to:

- ☒ the number of undecayed nuclei present at that instant
- ☐ the time since the decays started
- ☐ the time since the decay stoped
- ☐ none of these

91. As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75m from the origin. The z component of the angular momentum around the origin is:

- ☒ $6.0\text{kg} \cdot \text{m}^2/\text{s}$
- ☐ $9.0\text{kg} \cdot \text{m}^2/\text{s}$

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☐ 7.0kg . m2/s

☐ 8.0kg . m2/s

92. A body will be in translation equilibrium if:

☒ $F = 0$

☐ $F = 1$

☐ $F = 2$

☐ $F = 3$

93. The fundamental dimensions of angular momentum are .

☒ ML^2T^{-1}

☐ ML^3T^{-1}

☐ ML^4T^{-1}

☐ ML^5T^{-1}

94. The ultimate strength of a sample is the stress at which the sample

☐ begins to deform plastically.

☒ reaches its maximum resistance to applied force before failure.

☐ returns to its original shape after load removal.

☐ completely fractures without any prior deformation

95. The law of inertia was firstly formulated by:

☒ Newton

☐ Pascal

☐ Gillio

☐ Clouis

96. Sound waves are

☒ Longitudinal

☐ Infrasonic

☐ Audible

☐ Ultrafrasinic

97. One revolution is the same as:

☒ 2π rad

☐ 3π rad

☐ 4π rad

☐ 5π rad

OR

☐ 0.95 rad/s

☐ 1.57 rad/s

☒ 0.0524 rad/s

☐ 0.105 rad/s

98. A force of 100 N acts upon a body for five second. What will be the change in

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momentum?

- ☒ 500 NS
- ☐ 600 NS
- ☐ 700 NS
- ☐ 800 Ns

99. The dot product of vector A with itself is equal to

- ☒ A^2
- ☐ A^3
- ☐ A^4
- ☐ A^5

100. Swimming becomes possible because of:

- ☒ Third law of motion
- ☐ First law of motion
- ☐ Second law of motion
- ☐ None of these

101. Coulomb's law is only true for point charges whose sizes are

- ☒ Very small
- ☐ Very large
- ☐ Very big
- ☐ Very high

102. If the force acting on a body is doubled then the acceleration becomes:

- ☒ Double
- ☐ Single
- ☐ Triple
- ☐ fourth

103. The approximate value of g at an altitude above Earth equal to one Earth diameter is:

- ☒ 1.1 m/s^2
- ☐ 1.2 m/s^2
- ☐ 1.3 m/s^2
- ☐ 1.4 m/s^2

104. Young's modulus can be used to calculate the strain for a stress that is:

- ☒ Well below the yield strength
- ☐ Well above the yield strength
- ☐ Well down the yield strength
- ☐ Well up the yield strength

105. The rate of change of momentum of a body is equal to:

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☒ Applied force

☐ Un-applied force

☐ Applied velocity

☐ Un-applied velocity

106. As the wavelength of a wave in a uniform medium increases, its frequency will_____.

☒ Decrease

☐ Increase

☐ Less

☐ More

107. The momentum of an object at a given instant is independent of its:

☒ Acceleration

☐ Force

☐ Velocity

☐ Momentum

108. Work has dimensions like:

☒ Torque

☐ Force

☐ Velocity

☐ Acceleration

109. The unit $\text{kg.m}^2/\text{s}$ can be used for:

☒ Angular momentum

☐ Circular momentum

☐ None of these

☐ Both are correct

110. People try to keep the..... over their feet, in order to feel stable.

☒ Centre of gravity

☐ Final of gravity

☐ Initial of gravity

☐ None of these

111. A..... vector is obtained by dividing the vector by its magnitude:

☒ Unit

☐ Prefix

☐ Suffix

☐ Postfix

112. Which of the following is NOT one of Piaget's stages of development?

☒ Operational

☐ Non-operational

☐ Central

☐ Initial

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113. Which of the following process is suggested by Bandura that help human behavior to learn?

- ☒ Observation
- ☐ Experiment
- ☐ Conclusion
- ☐ Logical

114. All of the following are the areas of interest for Behaviorists EXCEPT:

- ☒ Unconscious desires
- ☐ Conscious desires
- ☐ Good desires
- ☐ Weak desires

115. Which of the following learning forms an association between two stimuli?

- ☐ Operant conditioning
- ☒ Classical conditioning
- ☐ Observational learning
- ☐ Latent learning

116. Helmholtz's explanation of color vision is called as what?

- ☐ Opponent-process theory
- ☒ Trichromatic theory
- ☐ Dual-process theory
- ☐ Retinex theory

117. In which of the following process sensory receptors become less sensitive to repeated presentations of the same stimulus?

- ☒ Sensory adaptation
- ☐ Habituation
- ☐ Perception
- ☐ Sensation

118. According to me here is two option correct but You select according to you Farha is interested to study about progressive change in individuals' body and their functions throughout the life span. She is more likely to study:

- ☐ Cognitive development
- ☐ Social development
- ☒ Biological changes
- ☐ Emotional development

119. All of the following are considered as the founders of gestalt school of thought EXCEPT:

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☒ Max Wertheimer

☐ Kurt Koffka

☐ Wolfgang Köhler

☐ John Watson

120. Which of the following is NOT a part of mid brain?

☐ Tectum

☐ Tegmentum

☒ Cerebrum

121. Superior colliculus Which of the following process is suggested by Bandura that help human behavior to learn?

☐ Imitation

☒ Observation

☐ Reinforcement

☐ Conditioning

122. Which of the following hormone is involved in the growth and proper functioning of gonads and ovaries?

☒ Gonadotropic hormone

☐ Thyroxine

☐ Insulin

☐ Adrenaline

123. Which of the followings are the branch-like structures that receive messages from other neurons?

☐ Axons

☒ Dendrites

☐ Cell body

☐ Myelin sheath

124. Who among the following talked about the “cognitive maps”?

☐ B.F. Skinner

☐ Edward Thorndike

☒ Tolman

☐ John Watson

125. Coherent sources of light mean ____.

☒ They must have the same frequency

☐ They must have different wavelengths

☐ They must have the same intensity

☐ They must be in phase with each other

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126. The focal length of a spherical mirror is N times its radius of curvature where N is:

- ☒ 1/2
- ☐ 1/4
- ☐ 2
- ☐ 1

127. The torque about any two points has the same value when the body is in.

- ☐ Rotational equilibrium
- ☐ Translational equilibrium
- ☐ Dynamic equilibrium
- ☒ Static equilibrium

128. A light entering into glass prism from air does not give change in its:

- ☒ Frequency
- ☐ Speed
- ☐ Wavelength
- ☐ Direction

129. The wavelength of red light is 700 nm. Its frequency is ____.

- ☐ 3.00×10^{14} Hertz
- ☒ 4.29×10^{14} Hertz
- ☐ 5.00×10^{14} Hertz
- ☐ 6.00×10^{14} Hertz

130. Bending of light around the edges of an obstacle called:

- ☐ Refraction
- ☐ Reflection
- ☒ Diffraction
- ☐ Dispersion

131. If I_0 is the peak value of an AC supply, then its rms value is given as I_{rms} :

- ☒ $\frac{I_0}{2}$
- ☐ $\frac{I_0}{\sqrt{2}}$
- ☐ $\sqrt{2} \cdot I_0$
- ☐ I_0

132. The total energy in an LC circuit is 5.0×10^{-6} J. If $L = 25$ mH the maximum current is:

- ☐ 10 mA
- ☒ 20 mA
- ☐ 30 mA

133. 40 m frequency of 500 Hz emits waves of wavelength 0.4 m, how long does the

ÂL-JÛÑÂÎD TÊCH

wave take to travel 600 m?

- ☐ 1.5 s
- ☐ 2 s
- ☒ 3 s
- ☐ 4 s

134. The center of mass of a uniform disk of radius R is located:

- ☐ At the edge of the disk
- ☐ At a point one-quarter of the radius from the center
- ☒ At the center
- ☐ At a point one-half of the radius from the center

135. Coulomb's law is only true for point charges whose sizes are

- ☐ Large
- ☒ Small
- ☐ Zero
- ☐ Finite but large

136. Newton-second is SI unit of:

- ☒ Impulse
- ☐ Pascal
- ☐ Newton
- ☐ Newton per second

137. The area under the curve of force-displacement graph represents:

- ☒ Fs
- ☐ As
- ☐ Ds
- ☐ Ms

138. When you step on the accelerator to increase the speed of your car, the force that accelerates the car is?

- ☒ Friction
- ☐ Motion
- ☐ Speed
- ☐ Velocity

139. Is the total momentum of a system is changing?

- ☒ A next external force must be acting on the system
- ☐ A next internal force must be acting on the system
- ☐ A next external force must not be acting on the system
- ☐ A next internal force must not be acting on the system

140. If the acceleration of a body is not uniform, then velocity-time graph will be:

- ☒ A curved line
- ☐ A straight line
- ☐ A triangle line
- ☐ A linear line

ÂL-JÛÑÂÎD TÊCH

141. The ability of a capacitor to store charge depends upon
- ☐ Type of dielectric used
 - ☐ Area of plates
 - ☒ All of these
 - ☐ Distance between plates
142. When you step on the accelerator to increase the speed of your car. The force that accelerates the car is
- ☒ The force of friction of the road on the tires
 - ☐ The force of your foot on the accelerator
 - ☐ The normal force of the road on the tires
 - ☐ The force of the engine on the drive shaft
143. Newton first law is also called:
- ☐ Law of torque
 - ☐ Law of gravitation
 - ☐ Law of force
 - ☒ Law of inertia
144. The Slope of velocity time graph shows the:
- ☒ Average acceleration of body
 - ☐ Average force of acting on body
 - ☐ Total work done on the body
 - ☐ Total distance covered by the body
145. An ultrasonic wave is sent from a ship towards the bottom of the sea. It is found that the time interval between the sending and in the seawater is 1400 m/s ?
- ☒ 1120 m
 - ☐ 560 m
 - ☐ 1400 m
 - ☐ 112 m
146. One newton is a force that produces an acceleration of 0.5 m/s in a body of mass:
- ☐ 1 kgf
 - ☐ 8 kg
 - ☐ 4 kg
 - ☒ 2 kg
147. A force of 120 N is exerted on a 40 kg container which sits on a floor. If the frictional force
- ☐ 5 ms^{-2}
 - ☐ 3 ms^{-2}
 - ☒ 1 ms^{-2}
 - ☐ 2 ms^{-2}
148. The speed of sound wave is determined by :

ÂL-JÛÑÂÎD TÊCH

- ☐ Number of harmonics present
- ☐ Its intensity
- ☐ Its amplitude
- ☒ The transmitting medium

149. A radian is about:

- ☐ 45°
- ☒ 57°
- ☐ 25°
- ☐ 37°

150. Force is a:

- ☒ Derived quantity
- ☐ Universal quantity
- ☐ Scalar quantity
- ☐ Base quantity

151. A municipal water supply is provided by a tall water tower. Water from this tower flows to a building. How does the water flow identical faucet on the second floor of the building?

- ☐ Water flows more rapidly out of the second-floor faucet.
- ☒ Water flows more rapidly out of the ground-floor faucet.
- ☐ Water flows at same speed out of both faucets.
- ☐ The speed of the water flows cannot be determined unless the height of the water tower is known.

152. Which of the following statement are TRUE of sound waves? Identify all that apply.

- ☐ To hear the sound of a tuning fork. The tines of the fork must move air from the fork to one's ear
- ☒ A sound wave is a mechanical wave
- ☐ A sound wave is a transverse wave
- ☐ Sound can travel through a vacuum

153. 0.0003711 in scientific notation is:

- ☐ 3.7×10^{-3}
- ☐ 3.7×10^{-1}
- ☐ 3.7×10^{-5}
- ☒ 3.7×10^{-4}

154. The newton second is a unit of :

- ☒ Linear momentum
- ☐ Angular momentum
- ☐ Work
- ☐ Power

ÂL-JÛÑÂÎD TÊCH

155. The physical quantity. Which oscillates in most wave is :
☐ Energy
☒ Amplitude
☐ Mass
☐ Wavelength
156. Sound and light wave both:
☐ Have similar wavelength
☒ Obey the law of reflection
☐ Travel through vacuum
☐ Travel as longitudinal wave
157. Which of the following stage refers to the transition between wakefulness and sleep?
☐ Stage-2 sleep
☒ Stage-1 sleep
☐ Stage-4 sleep
☐ Stage-3 sleep
158. Which of the following is true of all virtual images?
☒ They can be seen but not photographed
☐ They are smaller than the objects
☐ They are larger than the objects
☐ None of these
159. For an electron , the rest mass energy is
☐ 0.411 MeV
☒ 0.511 MeV
☐ 0.611 MeV
☐ 0.711 MeV
160. A transparent refracting medium bounded by the two curved surfaces called:
☐ Prism
☐ Mirror
☐ Glass
☒ Lens
161. There is a temperature at which the reading on the Kelvin scale is numerically:
☒ Equal to that on the Celsius scale
☐ Lower than that on the Celsius scale
☐ Equal to that on the Fahrenheit scale
☐ Less than zero
162. Which one of is in the order of decreasing frequency?

ÂL-JÛÑÂÎD TÊCH

☒ X-rays, radio waves, infrared rays

☐ Ultraviolet rays, visible light ,radio wave

☐ Infrared rays, visible light , X-rays

☐ Yellow, green, red

163. A particular device transmitted a signal in the form of electromagnetic wave of frequency 7.5×10^8 Hz . what is the wavelength of the signal?

☒ 0.4 m

☐ 40 m

☐ 4.0 m

☐ 4.04 m

164. A generator supplies 100V to the primary coil of a transformer . the primary has 50 turns the secondary voltage is :

☒ 1000V

☐ 500V

☐ 250V

☐ 100V

165. Polarization experiments provide evidence that light is:

☐ A longitude wave

☒ A transverse wave

☐ Some type of wave

☐ Nearly monochromatic

166. The units of the plank constant h are those of :

☒ Energy

☐ Angular momentum

☐ Momentum

☐ Frequency

167. We desire to make an LC current that ascllates 100 Hz using an inductance of 2.5H . we also need a capacitance of

☐ 1 F

☐ 1mF

☒ 1uf

☐ 100uf

168. If energy of photon is E, it is equvalent to mass:

☐ E/c^2

☐ Ec^2

☐ C^2/E

☒ Ec

169. The wave length of light beam A is twice the wavelength of light beam B. the

ÂL-JÛÑÂÎD TÊCH

energy of a photon in beam A is:

- ☒ Half the energy of photon in beam B
- ☐ One-fourth the energy of a photon in beam B
- ☐ Equal to the energy of a photon in beam B
- ☐ Twice the energy of photon in beam B

170. The reaction of an induction at 50 Hz is $10\ \Omega$, its reactance at 100Hz becomes:

- ☒ $20\ \Omega$
- ☐ $5\ \Omega$
- ☐ $2.5\ \Omega$
- ☐ $1\ \Omega$

171. The wavelength of red light is 700 nm. Its frequency is

- ☐ 4.29×10^3 Hertz
- ☐ 4.29×10^5 Hertz
- ☒ 4.29×10^4 Hertz OR 4.29×10^{14} Hertz
- ☐ 4.29×10^2 Hertz

172. $2d\sin\theta = m\lambda$ is called:

- ☐ Laplace's equation
- ☐ Reflection equation
- ☐ Refraction equation
- ☒ Bragg equation

173. A step-down transformer is used to :

- ☐ Increase the power
- ☐ Decrease the power
- ☐ Increase the voltage
- ☒ Decrease the voltage

174. The focal length of a spherical mirror is N times its radius of curvature where N is:

- ☐ $\frac{1}{4}$
- ☒ $\frac{1}{2}$
- ☐ 1
- ☐ 2

175. The appearance of colors in film is due to:

- ☒ Interference
- ☐ Diffraction
- ☐ Dispersion
- ☐ Polarization

176. Bending of light around the edges of an obstacle called:

- ☒ Refraction

ÂL-JÛÑÂÎD TÊCH

- ✚ Polarization
- ✚ Interference
- ✚ Diffraction

177. In constructing a thermometer it is NECESSARY to use a substance that:

- ✚ Expands linearly with rising temperature
- ✚ Will not freeze
- ✚ Will not boil
- ✚ Undergoes some change when heated or cooled

178. In Young's double slit experiment, the position of the bright fringe is given by:

- ✚ $Y_m = mU\lambda d$
- ✚ $Y_m = mU\gamma d$
- ✚ $Y_m = m\lambda d/L$
- ✚ $Y_m = mLd\lambda$

179. Constant-volume gas thermometers using different gases all indicate nearly the same temperature when in contact with the same object if :

- ✚ The volume are all extremely large
- ✚ The volume are all the same
- ✚ The pressure are all extremely large
- ✚ The particle concentration are all extremely small

180. In meta stable state electrons reside:

- ✚ 10^{-8} sec
- ✚ 10^{-5} sec
- ✚ 10^{-3} sec
- ✚ 10^{-13} sec

181. Following are the ways by which light can interact with matter , EXCEPT:

- ✚ Emission
- ✚ Interference
- ✚ Absorption
- ✚ Transmission

182. In RL series circuit phase angle is given by:

- ✚ $\tan^{-1}(\omega UR)$
- ✚ $\tan^{-1}(R/\omega L)$
- ✚ $\tan^{-1}(\omega LR)$
- ✚ $\tan^{-1}(\omega R/L)$

183. The theoretical upper limit for the frequency of electromagnetic wave is :

- ✚ Just slightly greater than of red light
- ✚ Just slightly less than that blue light
- ✚ The greatest x-ray frequency

ÂL-JÛÑÂÎD TÊCH

✚ There is no upper limit

184. The plants look green. It is because they have chlorophyll in them .chlorophyll make leaves green to absorb_____:

- ✚ Green Only
- ✚ Red only
- ✚ Blue only
- ✚ Red and blue both

185. Electric energy is measured :

- ✚ Kilowatt hour
- ✚ Kilowatt
- ✚ Horsepower
- ✚ Watt

186. The special theory of relativity is applicable to the object moving with maximum velocity equal to:

- ✚ Speed of light
- ✚ More than speed of light
- ✚ Less than speed of light
- ✚ Double the speed of light

187. In some movies you sometimes see an actor looking in a mirror and you can see his face in the mirror . during the filming of the scene what does the actor see in the mirror?

- ✚ His face
- ✚ Your face
- ✚ The movies cameras
- ✚ The director's face

188. The units of Stefan boltzomen constant are_____:

- ✚ $Wm^{-2}K^{-4}$
- ✚ $W^{-.2}m^{-.2}K^{-.4}$
- ✚ $Wm^{-.2}Kc^{-.4}$
- ✚ $Wm^{-.2}K^{-4}$

189. Two sources of light are said to be coherent if the waves produced by them have the same:

- ✚ Wavelength
- ✚ Amplitude
- ✚ Wavelength and amplitude
- ✚ Wavelength and constant phase difference

190. Antiparticle of electron is:

- ✚ Proton

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-  Photon
-  Neutron
-  Positron

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